Holonomy-Aware Attention via Discrete Fiber Bundle Geometry

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Abstract

We propose a geometric formulation of attention in which token representations reside in local vector spaces connected by learned parallel transport operators. In contrast to standard attention, which assumes a globally shared representation space, relational comparisons in our model are performed only after transporting representations into a common fiber. This construction induces a discrete fiber bundle structure over the token interaction graph. Within this framework, compositions of transport operators along loops define holonomy, which provides a natural geometric measure of contextual inconsistency and path dependence in relational comparisons. We introduce a holonomy-aware attention mechanism that integrates these ideas into a transformer-style architecture, and we prove that it reduces to standard attention exactly when the connection is flat, that its relational comparisons are gauge invariant, and that nontrivial holonomy induces genuine path dependence. We then prove a representational separation: a single standard attention layer with a position-free symmetric read-out is exactly permutation-invariant and therefore cannot represent a label defined by the ordered product of per-token transformations around a loop; for uniformly (Haar) distributed edges we further prove *subset blindness* any classifier reading a strict subset of the edges attains accuracy exactly $1/2$ so the shortcut-free property is a theorem rather than an empirical check. On an ordered state-tracking task with dense per-step labels we prove exact, enumerated accuracy ceilings for all order-blind predictors (0.488; 0.601 when granted the current move) and show the holonomy-aware layer exceeds them (0.971), training from task labels alone and extrapolating to sequences $8\times$ its training length. We show, within the attention layer itself and via a parameter-matched ablation, that the loop composition is *load-bearing*: a causal arm in which the closing edge enters the output is consistently more accurate than an otherwise identical inert arm in which it does not, and the gap grows as the learned transports become accurate, while a gauge-invariant read-out confirms the learned holonomy separates the classes. All of these results hold when the training signal constrains the transports. We are explicit about the boundary of the claim: we find that the same loop composition is *not* discovered from the thin binary task label alone, and we report this trainability limitation honestly; the state-tracking result shows the limitation is specific to thin labels and disappears under dense per-step supervision. We make no claim that any naturally occurring dataset carries non-trivial holonomy; identifying domains and observables in which such structure is genuinely present, and achieving end-to-end trainability from thin labels, are left as open problems.

1 Introduction

Attention mechanisms have become a central component of modern machine learning architectures, particularly in transformer-based models for sequence processing. In the standard formulation, relational comparisons between tokens are performed through inner products between query and key vectors in a shared representation space. This allows each token to interact with all others, enabling flexible modeling of long-range dependencies.

However, this formulation implicitly assumes that all token representations can be directly compared within a single globally aligned vector space. While effective in many settings, this assumption restricts the model to relational structures that can be consistently embedded within a single coordinate system.

Our central question is whether relational structure is better modeled not by enforcing global alignment, but by explicitly representing incompatibility between local coordinate systems.

In many systems, information is organized across multiple contexts that cannot be simultaneously reconciled within a single global representation. Locally consistent observations may become incompatible when combined globally, a phenomenon often referred to as contextual inconsistency. This arises in domains such as multi-perspective reasoning, contextual probabilistic models, and quantum contextuality.

In this work, we introduce a geometric formulation of attention in which token representations are treated as elements of local vector spaces connected by learned transport operators. Rather than assuming a shared representation space, relational comparisons are performed only after transporting representations into a common fiber. In this way, relational similarity depends not only on token content but also on the structure of the transport connecting them.

This construction naturally leads to a discrete bundle structure over the token graph. Within this framework, transporting representations along different paths may yield different results, giving rise to holonomy. Nontrivial holonomy provides a geometric signal of contextual inconsistency and induces path-dependent, context-sensitive relational comparisons.

Building on this perspective, we propose a holonomy-aware attention mechanism in which relational comparisons depend on learned transport operators between token representations. This extends standard attention by allowing interactions to depend on the geometry of the underlying connection, rather than assuming global alignment.

The main contributions of this work are as follows:

- We introduce a geometric formulation of attention in which token representations reside in local vector spaces connected by learned transport operators.
- We show that relational comparison under parallel transport naturally induces holonomy, providing a representation of contextual inconsistency in relational structures.
- We develop a transport-based attention mechanism that integrates these geometric ideas into a transformer-style architecture, and we establish its basic theoretical properties: exact reduction to standard attention under flat connections, gauge invariance of relational comparisons, and path dependence under nontrivial holonomy.
- We prove a representational lower bound: a single standard attention layer with a position-free symmetric read-out is exactly permutation-invariant and therefore cannot represent an ordered-product (holonomy) label; for Haar-distributed edges we prove that every classifier reading a strict subset of the edges is exactly at chance. The accuracy of the full-view baseline is assessed empirically, and the multi-layer case is treated as a conditional cost statement.
- We introduce an ordered state-tracking task with dense per-step labels and prove exact, enumerated accuracy ceilings for all order-blind predictors; the holonomy-aware layer exceeds these ceilings while training from task labels alone, and extrapolates to sequences $8\times$ its training length resolving the trainability limitation below in the dense-supervision regime.
- We show, via a parameter-matched ablation within the attention layer, that the loop composition is load-bearing: a causal arm (closing edge enters the output) consistently outperforms an

otherwise identical inert arm, with the gap growing as the learned transports become accurate. We are explicit that this holds under transport-level supervision, and we report honestly that the same structure is not discovered from task labels alone.

- We discuss directions for future work including multi-perspective reasoning, quantum contextuality, and probabilistic and financial contextuality while being explicit that whether genuine holonomy is present in such real-world data, and in which observables, remains an open empirical question.

Together, these results provide a perspective on attention mechanisms in which relational computation is governed not only by similarity of representations but also by the geometry of the transport structure connecting them.

2 Background

2.1 Transformers and Attention

Transformer architectures represent sequences using a stack of attention layers that allow each token to interact with all other tokens in the sequence. Given token representations

$$x_1, x_2, \ldots, x_L \in \mathbb{R}^{d_{\text{model}}},$$

attention mechanisms compute new representations by comparing tokens through learned query, key, and value projections.

For each token i, query, key, and value vectors are defined as

$$q_i = W_Q x_i, \qquad k_i = W_K x_i, \qquad v_i = W_V x_i,$$

where W_Q, W_K, W_V are learned linear maps.

The attention score between tokens i and j is then computed using a scaled dot product,

$$s_{ij} = \frac{\langle q_i, k_j \rangle}{\sqrt{d}},$$

and attention weights are obtained by applying a softmax normalization

$$\alpha_{ij} = \frac{\exp(s_{ij})}{\sum_{j'} \exp(s_{ij'})}.$$

The output representation at token i is given by the weighted aggregation

$$h_i = \sum_j \alpha_{ij} v_j.$$

A central feature of this formulation is that all token representations are compared within a single shared vector space. In other words, relational similarity between tokens is evaluated directly through inner products in a global representation space.

The geometric formulation introduced in this work modifies this assumption. Instead of requiring token representations to lie in a single shared space, each token is associated with its own local representation space, and relational comparison is performed only after representations are aligned through parallel transport.

2.2 Discrete Fiber Bundles on Graphs

We model a token sequence together with its admissible interactions by a graph

$$G = (\mathcal{V}, \mathcal{E}),$$

where each node $i \in \mathcal{V}$ corresponds to a token and each edge $(i, j) \in \mathcal{E}$ indicates that information may be directly compared or transported between tokens i and j. While sequential data naturally induces a chain, we allow more general graph structure so that nontrivial loops and relational cycles may be represented.

We regard G as a one-dimensional simplicial complex whose 0-simplices are the nodes $\mathcal{V}$ and whose 1-simplices are the edges $\mathcal{E}$. Its first homology group $H_1(G, \mathbb{Z}) \cong \mathbb{Z}^{\beta_1}$ classifies independent cycles, where

$$\beta_1 = |\mathcal{E}| - |\mathcal{V}| + 1$$

is the cycle rank of G. Every loop in G can be expressed as a combination of a fixed basis of β_1 cycles, obtained by choosing any spanning tree of G and taking the unique cycle induced by each non-tree edge.

To each node $i \in \mathcal{V}$ we attach a d-dimensional vector space

$$\mathcal{F}_i \cong \mathbb{R}^d,$$

which we interpret as the local representation space, or fiber, at node i. Thus, token features are not assumed a priori to live in a single global ambient space; instead, each token is represented in its own local coordinate system.

For each edge $(i, j) \in \mathcal{E}$, we define a parallel transport operator

$$U_{ij} : \mathcal{F}_j \to \mathcal{F}_i,$$

which maps representations from the fiber at node j into the fiber at node i. In this work we assume

$$U_{ij} \in \mathrm{SO}(d),$$

so that transport preserves inner products and norms. When reverse transport is defined, we require

$$U_{ji} = U_{ij}^{-1} = U_{ij}^{\top}.$$

The collection $\{\mathcal{F}_i, U_{ij}\}_{(i,j)\in\mathcal{E}}$ defines a discrete vector bundle with orthogonal connection over the graph G. Similar constructions arise in lattice gauge theory, where group-valued link variables assigned to lattice edges represent parallel transport between neighboring sites [19, 15].

Given a path

$$\pi = (i_0, i_1, \ldots, i_k),$$

its associated transport operator is the ordered composition

$$U_\pi = U_{i_0 i_1} U_{i_1 i_2} \cdots U_{i_{k-1} i_k},$$

which maps vectors from $\mathcal{F}_{i_k}$ to $\mathcal{F}_{i_0}$. Thus, if $x \in \mathcal{F}_{i_k}$, then $U_\pi x \in \mathcal{F}_{i_0}$ is the result of transporting x along the path π.

This discrete bundle formulation provides the basic geometric structure used throughout the paper. Rather than comparing token representations directly in a single shared space, we compare them only after transport into a common fiber. In this way, relational comparison depends not only on token content but also on the connection structure defined over the token graph.

A globally consistent representation space is recovered only when the local fibers can be simultaneously aligned by a family of orthogonal changes of basis. In general, such a global alignment need not exist, and the resulting obstruction is captured by holonomy, introduced in the next subsection.

2.3 Holonomy, Contextuality, and Sheaf-Theoretic Consistency

A central motivation for the proposed architecture is the notion of *contextuality*, where locally compatible observations cannot always be globally reconciled. In the sheaf-theoretic formulation of contextuality introduced by Abramsky and Brandenburger, contextuality arises when locally consistent assignments of measurement outcomes cannot be extended to a single globally consistent assignment across all contexts [1]. In this setting, contexts correspond to sets of jointly measurable observables, and incompatibility between contexts appears as an obstruction to constructing a global section of the associated presheaf.

The geometric viewpoint adopted in the present work is closely related but emphasizes transport structure rather than measurement assignments. Following the perspective developed by Magnot [13], contextuality can be interpreted geometrically through path dependence in a relational transport structure. Parallel transport along different relational paths may produce inconsistent transformations, and nontrivial holonomy provides a natural geometric signal of this obstruction to global consistency.

In a discrete fiber bundle over a graph, each node i carries a vector space $V_i \cong \mathbb{R}^d$, and edges are equipped with parallel transport operators

$$U_{ij} : V_j \to V_i,$$

which we assume to lie in the orthogonal group $\mathrm{SO}(d)$.

Parallel transport along a path $\pi = (i_0, i_1, \ldots, i_k)$ is given by the ordered product

$$U_\pi = U_{i_0 i_1} U_{i_1 i_2} \cdots U_{i_{k-1} i_k}.$$

When π forms a loop, i.e., $i_k = i_0$, the resulting operator

$$H_\pi = U_{i_0 i_1} U_{i_1 i_2} \cdots U_{i_{k-1} i_0}$$

is called the *holonomy* associated with that loop.

Holonomy measures the failure of representations to return unchanged after transport around a closed path. When all loop holonomies are equal to the identity, the bundle is flat in the discrete sense, meaning that a global frame exists in which all representations can be consistently compared.

This perspective connects naturally with sheaf-theoretic notions of contextuality. In that setting, nodes may be interpreted as local contexts and compatibility conditions describe how local representations should agree on overlaps. Nontrivial holonomy then acts as an obstruction to the existence of a globally consistent choice of representation, analogous to the absence of a global section satisfying all local compatibility constraints.

Motivated by this viewpoint, we interpret contextual inconsistency in relational representation learning as the presence of nontrivial holonomy in the token interaction graph. In the architecture introduced here, this geometric obstruction induces path-dependent relational comparison, allowing attention to encode context-sensitive interactions between tokens.

Holonomy has also been connected to the notion of *memory effects* in gauge theory and general relativity[16]. From this viewpoint, holonomy captures the accumulation of path-dependent transformations and therefore provides a geometric representation of memory. In the context of the present work, nontrivial holonomy in the discrete transport structure plays a similar conceptual role: relational comparisons between token representations depend on the sequence of transports through which information propagates, thereby introducing a form of geometric memory into the attention mechanism. Recent work by Sung [18] develops this memory interpretation in a complementary direction, using non-abelian holonomy on $\mathrm{SO}(N)$ to protect the *sequential* hidden-state evolution

of a single recurrent model against noise. We contrast that setting with ours in section 10: there holonomy protects memory *along the time axis of one model*, whereas here holonomy measures inconsistency *across the relational graph connecting distinct representations*.

3 Transport-Based Attention

We now introduce the proposed attention mechanism built on the discrete bundle structure defined in the previous section. The key idea is that token representations are first embedded in a common ambient feature space and then projected into local fiber spaces associated with each token. Relational comparison between tokens is performed only after representations have been transported into a common fiber.

Let $x_i \in \mathbb{R}^D$ denote the ambient representation of token i. To obtain fiber representations, we apply a shared linear projection

$$f_i = W_F x_i,$$

where $f_i \in \mathcal{F}_i$ and $\mathcal{F}_i \cong \mathbb{R}^d$ denotes the local fiber attached to token i.

As in standard transformer architectures, we compute query, key, and value representations from the fiber vectors using shared linear projections

$$q_i = W_Q f_i, \qquad k_i = W_K f_i, \qquad v_i = W_V f_i,$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d\times d}$.

Since q_i, k_i, and v_i are elements of the fiber $\mathcal{F}_i$, relational comparison between tokens i and j requires transporting one representation into the other's fiber. Using the parallel transport operator $U_{ij} : \mathcal{F}_j \to \mathcal{F}_i$, we define the transported key

$$\tilde{k}_{j\to i} = U_{ij} k_j.$$

The attention score between tokens i and j is then computed as

$$s_{ij} = \frac{q_i^\top (U_{ij} k_j)}{\sqrt{d}}.$$

Attention weights are obtained using the softmax normalization

$$\alpha_{ij} = \frac{\exp(s_{ij})}{\sum_{j'} \exp(s_{ij'})}.$$

To aggregate information from neighboring tokens, value vectors are also transported into the fiber at node i

$$\tilde{v}_{j\to i} = U_{ij} v_j.$$

The updated representation at node i is then

$$h_i = \sum_j \alpha_{ij} (U_{ij} v_j),$$

where $h_i \in \mathcal{F}_i$ remains an element of the fiber at token i. The result may subsequently be mapped back to the ambient representation space through a standard output projection.

Thus, both similarity computation and value aggregation occur only after representations have been aligned through parallel transport. In contrast to standard attention, where all tokens are

compared within a single shared representation space, the proposed mechanism performs relational comparison within local fibers connected by the discrete bundle structure.

To illustrate the role of local gauges, consider a closed path through five nodes

$$1 \to 2 \to 3 \to 4 \to 5 \to 1.$$

If transport operators are written in the gauge-covariant form

$$T_{ij} = R_i G_{ij} R_j^\top,$$

then the loop transport obtained by composing operators along the path is

$$H = T_{15} T_{54} T_{43} T_{32} T_{21}.$$

Substituting the decomposition gives

$$H = (R_1 G_{15} R_5^\top)(R_5 G_{54} R_4^\top)(R_4 G_{43} R_3^\top)(R_3 G_{32} R_2^\top)(R_2 G_{21} R_1^\top).$$

Since $R_i^\top R_i = I$, all intermediate gauge factors cancel pairwise, yielding

$$H = R_1 (G_{15} G_{54} G_{43} G_{32} G_{21}) R_1^\top.$$

Thus the holonomy of the loop depends only on the intrinsic transports G_{ij} up to conjugation by the starting gauge R_1. Quantities such as the eigenvalues or trace of H are therefore gauge-invariant.

4 Holonomy and Inconsistency

We now formalize how holonomy quantifies inconsistency in relational comparisons induced by transport.

Consider a loop

$$\gamma = (i_0, i_1, \ldots, i_k = i_0)$$

in the token graph. The associated holonomy operator is

$$H_\gamma = U_{i_0 i_1} U_{i_1 i_2} \cdots U_{i_{k-1} i_0}.$$

To measure deviation from consistency, we define the holonomy defect

$$\mathrm{inc}(\gamma) = \|H_\gamma - I\|_F,$$

where $\|\cdot\|_F$ denotes the Frobenius norm.

A nonzero holonomy defect indicates that transport around the loop depends on the path taken. Consequently, relational comparisons cannot be globally aligned, and attention scores may vary depending on the sequence of intermediate relations.

Remark 1 (Basis loops and gauge-covariance of holonomy)**.** Because holonomy defines a group homomorphism

$$\mathrm{Hol}\colon H_1(G, \mathbb{Z}) \longrightarrow \mathrm{SO}(d), \qquad \gamma \longmapsto H_\gamma,$$

satisfying $H_{\gamma_1 \cdot \gamma_2} = H_{\gamma_1} H_{\gamma_2}$, it suffices to evaluate $\mathrm{inc}(\gamma)$ on the β_1 basis loops determined by the non-tree edges of any spanning tree of G. The holonomy of every other loop follows by composition,

so the complete contextual inconsistency of the connection is captured by exactly $\beta_1 = |\mathcal{E}| - |\mathcal{V}| + 1$ scalar values.

It is important to note that the holonomy matrix H_γ itself is *not* gauge-invariant: under a local gauge transformation $\{R_i\}_{i\in\mathcal{V}}$ it transforms by conjugation, $H_\gamma \mapsto R_{i_0} H_\gamma R_{i_0}^{-1}$, where i_0 is the basepoint of γ. This conjugation arises because the intermediate gauge rotations cancel pairwise around the loop, but the outermost factor R_{i_0} appears on both the left and the right with no partner to absorb it. Since $\mathrm{SO}(d)$ is non-abelian for $d \geq 3$, conjugation is in general nontrivial and H_γ changes with the choice of local frame. The holonomy defect $\mathrm{inc}(\gamma) = \|H_\gamma - I\|_F$, however, is fully gauge-invariant: the Frobenius norm satisfies $\|RXR^\top\|_F = \|X\|_F$ for any orthogonal R, so $\mathrm{inc}(\gamma)$ depends only on the intrinsic geometry of the connection and not on the choice of local coordinates.

4.1 Where inconsistency can reside: a telescoping obstruction

A structural observation guides the analysis that follows, and it constrains how one must model any candidate application. Suppose each node carries a single vector $\sigma_i \in \mathcal{F}_i$ and the transport on each edge is defined as the (minimal) rotation aligning one node's vector onto its neighbor's. Then the loop holonomy *telescopes to the identity*: if $U_{i\,i+1}$ is chosen so that $U_{i\,i+1}\sigma_{i+1} = \sigma_i$ and, for orthonormal frames, $U_{i\,i+1} = \sigma_i \sigma_{i+1}^\top$ along the loop, the ordered product collapses as

$$H_\gamma = (\sigma_{i_0}\sigma_{i_1}^\top)(\sigma_{i_1}\sigma_{i_2}^\top)\cdots(\sigma_{i_{k-1}}\sigma_{i_0}^\top) = \sigma_{i_0}\sigma_{i_0}^\top,$$

which reduces to the identity on the relevant subspace, giving $\mathrm{inc}(\gamma) = 0$ regardless of the node contents. Consequently, *genuine holonomy cannot be carried by node content under pure frame-alignment transport*: it must reside in the *relational structure* the transport operators themselves whose loop product need not admit a consistent global assignment of node frames. This is the discrete analogue of the statement that a connection may have no global flat section. It motivates the load-bearing analysis of section 6: we prove that standard attention cannot represent an ordered-product label, and then show on a shortcut-free instance that the holonomy-aware layer's performance depends causally on loop closure.

5 Theoretical Properties

In this section we describe several structural properties of the proposed transport-based attention mechanism. These results clarify the relationship between our formulation and standard transformer attention and illustrate how geometric properties of the connection influence relational comparison. Formal proofs are provided in the supplementary material.

5.1 Reduction to Standard Attention under Flat Connections

The proposed mechanism generalizes standard attention and reduces to it when the connection admits a global alignment.

Theorem 1 (Reduction to Standard Attention)**.** *Suppose the connection $\{U_{ij}\}$ is flat, meaning that there exist orthogonal matrices $R_i \in SO(d)$ such that*

$$U_{ij} = R_i R_j^{-1}.$$

Define transformed representations

$$\tilde{x}_i = R_i^{-1} x_i.$$

Then transport-based attention applied to $\{x_i\}$ is equivalent to standard dot-product attention applied to the transformed representations $\{\tilde{x}_i\}$.

Remark 2. Theorem 5.1 shows that standard attention corresponds to the special case in which token representations can be globally aligned within a single shared vector space. Equivalently, flatness means that the holonomy representation $\mathrm{Hol}\colon H_1(G,\mathbb{Z}) \to \mathrm{SO}(d)$ is trivial — every basis loop has holonomy equal to the identity. In contrast, the proposed model allows the possibility that no such global alignment exists. In this case, token representations remain locally defined within their respective fibers, and relational comparisons must be mediated by the connection structure defined over the token graph.

5.2 Path Independence under Trivial Holonomy

The behavior of transport across the token graph depends on the holonomy of the connection.

Theorem 2 (Path Independence)**.** *If the holonomy associated with every loop in the token graph is equal to the identity, then parallel transport between any two nodes depends only on the endpoints and not on the chosen path.*

Remark 3. In this case, relational comparison between tokens becomes path independent. Transported attention scores depend only on the tokens being compared and not on the intermediate relational structure.

5.3 Gauge Invariance

The proposed attention mechanism is invariant under local orthogonal changes of basis in the fibers.

Theorem 3 (Gauge Invariance)**.** *Let $\{R_i\}_{i\in\mathcal{V}}$ be a family of orthogonal matrices with $R_i \in SO(d)$. Consider the transformations*

$$x_i \to R_i x_i,$$

$$U_{ij} \to R_i U_{ij} R_j^{-1}.$$

Then the attention scores s_{ij} and outputs h_i remain unchanged.

Remark 4. This invariance reflects the fact that the mechanism depends only on the geometric relations induced by the connection rather than on any particular choice of local coordinates.

5.4 Context Sensitivity under Nontrivial Holonomy

When the connection exhibits nontrivial holonomy, relational comparison becomes sensitive to the structure of paths in the token graph.

Proposition 1. *If the holonomy associated with a loop γ is nontrivial, then transporting a representation around γ changes the orientation of vectors in the corresponding fiber. Consequently, relational comparisons involving transported representations may depend on the path taken through the graph.*

Remark 5. This property allows the proposed attention mechanism to encode context-sensitive relational structure. Nontrivial holonomy introduces a geometric form of path dependence, enabling the model to represent interactions that cannot be globally aligned in a single representation space.

6 Is Loop Composition Load-Bearing in the Attention Layer?

The theoretical results above establish what the mechanism *represents*. We now ask an empirical question that can be answered cleanly and without appeal to any real-world dataset: *within the holonomy-aware attention layer itself, does the mechanism's output actually depend on the closing of the loop is the loop composition load-bearing, or does the layer succeed by routing information around the loop structure?* We answer this with a representational lower bound for standard attention, a controlled shortcut-free detection task, and a matched-parameter ablation. We make no claim of discovery in naturally occurring data; that we discuss only as future work in section 9.

6.1 Load-bearing holonomy in the attention layer

We ask whether the loop composition is *load-bearing within the attention mechanism itself*: does the mechanism's output actually depend on the closing of the loop, or could it succeed by routing information around the loop structure? We answer this in three steps. First, we prove a representational lower bound: a single standard attention layer with a symmetric read-out cannot represent a label defined by the ordered loop product (section 6.1). Second, we exhibit a controlled, shortcut-free detection task whose label is exactly such an ordered product. Third, on that task we run a matched-parameter ablation isolating loop closure as the only difference between two arms.

A representational lower bound for standard attention. Before the ablation, we establish why the task is out of reach for ordinary attention. Consider a single standard self-attention head with linear q, k, v maps and softmax weights, followed by the read-out used in our baseline (residual add, mean-pool over positions, linear classifier), with no positional encoding.

Lemma 1 (Exact permutation invariance)**.** *For any width, any number of heads, and any parameter setting, the decision function of a single position-free standard attention layer followed by mean-pooling and a linear classifier is invariant under every permutation of its input tokens.*

Remark 6 (Structural intuition; carries no proof weight)**.** Softmax normalization couples all tokens and introduces interactions of unbounded polynomial order a third-order mixed finite difference of a softmax attention layer is nonzero, so the layer is provably *not* an order-≤ 2 functional, and we make no degree claim. The surviving structural observation is that every cross-token dependence in a single layer factors through pairwise query–key forms and per-token linear value maps; no operation contracts an index through a *chain* of tokens, which is what the ordered product $(G_{L-1})_{ab}(G_{L-2})_{bc}\cdots$ requires. The formal results rest solely on lemma 1. (Proof in section A.)

Proposition 2 (Standard attention cannot represent the loop product)**.** *Extend the label to all of* $\mathrm{SO}(d)^L$ *by* $\hat{y}(X) = \mathbf{1}[\,G_{L-1}\cdots G_0 = R_\pi\,]$*. There exist inputs related by a single transposition of two tokens on which* $\hat{y}$ *takes different values; hence, by lemma 1, no single position-free standard attention layer with symmetric read-out equals* $\hat{y}$*, at any width or head count.*

This is a statement about representability, and we are explicit about what it does not say: on the support of the task distribution the unordered multiset of edges almost surely determines the label (an ordering search detects whether some ordering composes to a π-rotation; section A), so permutation invariance alone does not cap on-support accuracy. Accuracy is instead controlled in two rigorous pieces. First, for the Haar variant of the edge distribution we prove *subset blindness* (theorem 5): the joint law of *any* strict subset of the edges is identical across the two classes, so every classifier of any architecture, at any width, trained or not that reads fewer than all L edges attains accuracy exactly $1/2$. Second, for the full-view softmax baseline no distribution-level theorem is

possible, since the complete configuration genuinely determines the label; its chance-level accuracy (section 6.1) is reported as a measurement. A parameter-matched multi-layer transformer is not ruled out: escaping proposition 2 requires either depth growing with L, or latent memory carrying the ordered partial products. Encoder–decoder architectures, including Transformers, fall within the locally-Lipschitz class of Anschuetz et al. [5] (their Theorem 3), with effective memory counted as model dimension times sequence length, so the latter route incurs their memory lower bound; we note their separation is worst-case (existence of inputs forcing an error) while our subset-blindness and ceiling results are distributional, so neither implies the other. Full statements and proofs are in section A.

Positional encodings do not circumvent the obstruction. A natural objection is that standard attention is rarely used without positional encodings, which break the permutation symmetry invoked above. The obstruction, however, is not merely permutation symmetry but the *commutativity of the aggregation*. As observed by Sung [18], standard attention combines values additively, $\sum_i \alpha_i V_i$; positional encodings including relative schemes such as RoPE and ALiBi break the permutation symmetry of the weights but are still consumed by a commutative sum, so sequence order is mapped into the geometry of the embedding space rather than represented as an explicit non-commutative composition of operators. We scope this precisely: *within a single layer with linear value maps*, the summands are linear images of individual tokens, so the summation itself performs no composition positional information changes *what* is summed but cannot make the sum compute the ordered product. Constructions that do carry order through a sum (e.g., pre-multiplying each summand by its ordered prefix) require the prefix products to already be computed, which is beyond a single layer; with depth, positionally encoded transformers can represent ordered products, which is why the multi-layer case is treated as a conditional cost statement above. Our baseline reflects this concretely: the standard model receives sinusoidal positional encodings in 16 of its 32 input dimensions, giving every position a distinct code, yet it remains at chance on the shortcut-free task. Positional encoding supplies the *order of appearance* of the tokens but not the ordered product the label requires. The holonomy-aware layer instead carries order structurally, through the composition of transport operators, and therefore uses no positional encoding.

A shortcut-free loop-detection task. We place $L = 6$ tokens on a ring. Each edge t carries a hidden rotation $G_t \in \mathrm{SO}(d)$ ($d = 4$); the ordered loop product $H = G_{L-1} \cdots G_1 G_0$ is either the identity (label 0) or a rotation by π in a fixed plane (label 1), the latter being the discriminating non-trivial holonomy. Each token is encoded only from the raw entries of its own edge rotation G_t; the individual rotations are otherwise random, so that no single edge reveals the label. We verified that the task admits no local or low-order shortcut: linear and nonlinear probes on individual edges, on the closing edge alone, and on all edges jointly all perform at chance (≈ 0.50), while the ordered loop product recovers the label exactly. The label is therefore a genuine property of the ordered composition around the loop and cannot be obtained without composing the transports in order. For the Haar variant of the edge distribution the strict-subset case is a theorem rather than a check: every read-out of fewer than all L edges is exactly at chance (theorem 5).

Two arms differing only in loop closure. We train the holonomy-aware attention layer in two modes that share *identical parameter counts and identical per-edge transports*, differing only in how transports are composed into the attention output. The CAUSAL arm uses arc transport, so that the closing edge enters the relational comparison and the loop holonomy influences the output (section 3). The INERT arm uses endpoint transport $C_i^\top C_j$, which telescopes: the closing edge never

	label accuracy		transport error	
λ	CAUSAL	INERT	CAUSAL	INERT
0 (labels only)	0.507	0.515	0.493	0.478
5	0.781	0.716	0.027	0.029
20	0.803	0.731	0.026	0.026
100	**0.818**	0.693	0.025	0.025

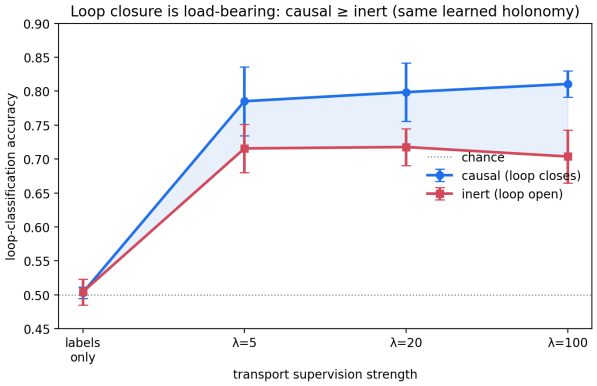
Table 1: Load-bearing ablation on the shortcut-free loop-detection task. The CAUSAL and INERT arms are parameter-matched and share identical per-edge transports; they differ only in whether the closing edge enters the output. The causal arm is consistently more accurate, and the gap grows from 0.065 at $\lambda = 5$ to 0.125 at $\lambda = 100$, where the causal arm improves while the inert arm regresses. The transport errors are nearly identical across arms, confirming the gap is due to loop closure rather than transport fidelity.

enters the output, so the loop holonomy is causally inert (section 4.1). Because the two arms are otherwise parameter-matched and receive the same transports, any performance gap is attributable to loop closure alone, not to capacity.

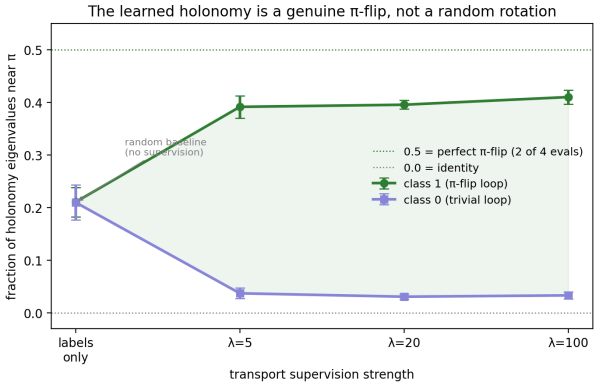
We observed that the loop composition is not discovered from the binary label alone by either arm an instance of the trainability difficulty of the full architecture, which we discuss in section 7. We therefore supply an auxiliary objective that supervises each model's per-edge transport toward the planted G_t, with weight λ; this constrains the transports directly. We report, for a range of λ, the transport error $\|G_t^{\text{model}} - G_t^{\star}\|$, the label accuracy, and a gauge-invariant read-out of the learned holonomy: the defect $\text{inc}(\gamma) = \|H_\gamma - I\|_F$ and the dominant eigenangle of H_γ, each averaged separately over the two classes. All entries are means over three seeds.

Loop closure is load-bearing, and its contribution grows with transport accuracy. As the auxiliary weight increases, the transport error falls sharply and label accuracy rises for both arms, but the CAUSAL arm is consistently more accurate than the parameter-matched INERT arm, and the gap *widens* as the transports become more accurate (table 1; figure 1). At the strongest supervision the causal arm continues to improve while the inert arm regresses, so that with accurate transports the loop closure accounts for a growing share of performance. Since the two arms are identical except for whether the closing edge enters the output, this establishes that the loop composition the holonomy is load-bearing in the attention mechanism, and is not an artifact of the additional transport parameters, which both arms possess in equal measure.

The learned holonomy separates the classes. The gauge-invariant read-out confirms that the discriminating quantity is the holonomy itself. Once the transports are accurate ($\lambda \geq 5$, transport error ≈ 0.025), the learned loop product separates the two classes as constructed: for label-1 samples the dominant eigenangle of H_γ concentrates near π (≈ 2.8 radians, against the planted value $\pi \approx 3.14$) and the defect $\text{inc}(\gamma)$ near the value 2.83 of an exact π-rotation, while for label-0 samples both quantities are substantially smaller and decrease further as accuracy improves. The class separation in the holonomy read-out (figure 1b) sharpens together with transport accuracy, so the loop product is demonstrably the carrier of the class signal, consistent with the ablation.



(a) Loop closure is load-bearing. The CAUSAL arm (loop closes) is consistently at least as accurate as the parameter-matched INERT arm (loop open); at labels-only ($\lambda = 0$) both sit at chance, and the gap favouring CAUSAL persists as transport supervision increases.



(b) The learned holonomy is a genuine π-flip, not a random rotation. As transport supervision increases, the fraction of holonomy eigenvalues near π separates the two classes: class 1 (π-flip loop) approaches the 0.5 value of an exact π-rotation (two of four eigenvalues at -1), while class 0 (trivial loop) collapses toward identity.

Figure 1: Load-bearing ablation on the shortcut-free loop-detection task, over transport-supervision strength λ (means over seeds; error bars show spread). **(a)** Loop closure is causally load-bearing: the CAUSAL and INERT arms are parameter-matched and share identical per-edge transports, differing only in whether the closing edge enters the output, yet CAUSAL is consistently more accurate. **(b)** A gauge-invariant read-out confirms the discriminating quantity is the holonomy itself: the learned loop product concentrates near a genuine π-flip for class 1 and near identity for class 0.

6.2 Interpretation and scope

Taken together, the lower bound and the ablation delineate a precise and honest claim. A single standard attention layer with a symmetric read-out provably cannot represent the ordered-product label (section 6.1); on a shortcut-free instance of exactly such a label, the holonomy-aware layer's performance depends causally on loop closure, under a matched-parameter ablation, with the contribution growing as the transports become accurate (section 6.1). This result holds in a specific regime: the training signal constrains the transports, here through an auxiliary transport objective. Establishing that loop closure is load-bearing under transport-level supervision is the contribution of this section; whether the same loop composition can be discovered end-to-end from task labels alone is a separate question, which we address next.

7 Trainability from Labels Alone, and Its Limits

The experiments of section 6 share a common ingredient: in each, the training signal directly constrains the transports. This is a genuine and useful regime it covers settings in which relational structure is observed, or in which intermediate supervision is available but it leaves open the sharper question of whether the holonomy-aware attention layer can *discover* the loop composition from task labels alone, with no supervision of the transports.

Our experiments indicate that, on the shortcut-free loop-detection task of section 6.1, it does not. With only the binary label as a training signal, neither the causal nor the inert arm rises above chance: the transports are not driven toward any consistent structure, and the ordered loop

product is not formed. The loop composition becomes learnable, and the causal advantage becomes visible, only once the auxiliary transport objective is supplied. We report this limitation plainly. It is consistent with the structural observation that a label defined by an ordered, non-commutative product of transports provides a distant and weakly-shaped gradient for the per-edge transports, and with the trainability difficulties that geometric and gauge-structured parameterizations are known to raise. It also matches our own observation that the full architecture exhibits seed-dependent variability when the transports are unsupervised.

We therefore state the scope of the empirical results without inflation. The mechanism *represents* holonomy exactly (section 5); the holonomic structure is *learnable and recoverable* under transport-level supervision (section 6); and within the attention layer the loop composition is *load-bearing* under a matched-parameter ablation (section 6.1). We do not claim that the mechanism discovers holonomy from task labels alone; on the contrary, we find that in our setting it does not, and we regard end-to-end trainability together with the identification of naturally occurring data in which above-floor holonomy is genuinely present as the central open problems for this framework. A faithful account of this boundary is more useful than an inflated one: it tells a practitioner exactly what the mechanism can be relied upon to do (recover and use holonomy when the transports are constrained) and what it cannot yet be relied upon to do (induce that structure from labels alone). Section 8 sharpens this boundary: the limitation is specific to the *thin* binary label, and disappears under dense per-step supervision.

8 Dense Supervision Restores Trainability: Ordered State Tracking

The trainability limitation of the previous section admits a sharp hypothesis: the failure is a property of the *training signal*, not of the mechanism. A single binary label at the end of a long non-commutative computation provides a distant, weakly-shaped gradient for the per-edge transports; a target at *every* step feeds every edge directly. We test this hypothesis on an ordered state-tracking task with dense per-step labels and no auxiliary transport supervision of any kind.

Task. Fix four generator rotations $g^{(1)}, \dots, g^{(4)} \in \mathrm{SO}(4)$, drawn once and frozen. A sample of length L is a uniformly random move sequence $m_1, \dots, m_L \in \{1, \dots, 4\}$; the state after step t is the running product $S_t = g^{(m_t)} \cdots g^{(m_1)}$, and the label at step t is a fixed lossy hash of S_t into 8 buckets (sign pattern of a frozen random projection). The model must output the label at every step. We verified the task is well-posed and shortcut-free by exhaustive enumeration: the same last move maps to many different labels (no last-move shortcut), the update rule is length-independent (states remain well-spread at all lengths tested), and at $L = 6$, 95.2% of move multisets map to more than one final label depending on order order is load-bearing.

An exact ceiling for order-blind prediction. The finite alphabet permits something the loop task cannot: an exact, information-theoretic ceiling computed by enumeration, binding *every* predictor limited to order-blind information.

Theorem 4 (Exact ceiling for summary-limited predictors)**.** *Fix a position t and let T be any summary statistic of the move prefix. Any predictor, deterministic or randomized, whose output depends on the prefix only through T has per-step accuracy at most* $\mathrm{Ceil}_t(T) = \mathbb{E}_T\big[\max_y \Pr(Y_t = y \mid T)\big]$*, attained by the majority vote on each value of T. With a finite alphabet,* $\mathrm{Ceil}_t(T)$ *is computed exactly by enumerating all 4^t prefixes.*

The proof (section B) is elementary conditioning plus the fact that a weighted average cannot exceed its largest term which is precisely its strength: the bound holds at any width, for any architecture, trained by any method. Enumerating all $4^8 = 65{,}536$ sequences of the actual task yields, for the dense metric (mean over positions 1..8):

predictor class	exact ceiling
know nothing (majority class)	0.214
any function of the move multiset (order-blind)	0.488
any function of (multiset, current move)	0.601
holonomy-aware layer (measured)	**0.971**

(The naïve uniform-chance figure $1/8 = 0.125$ understates the floor: the buckets are not uniform, and the exact majority-class baseline is 0.214.)

Setup and results. We train at $L_{\text{train}} = 8$ with per-step cross-entropy only no auxiliary loss and evaluate at $L \in \{8, 16, 32, 64\}$ (3 seeds). The holonomy-aware model uses no positional encoding (order is carried by the transports, with a causal mask); the standard baseline receives sinusoidal positional encodings *and* a causal mask strictly more positional machinery. Results (figure 2): the holonomy-aware layer reaches 0.971 dense accuracy at the training length and degrades gracefully under length extrapolation (0.87, 0.68, 0.49 at $L = 16, 32, 64$), while the standard baseline starts at 0.40 and collapses toward chance. Since $0.971 > 0.601$, theorem 4 certifies that the model exploits genuine order information: no order-blind mechanism, at any capacity, can reach that accuracy.

Scope. Three boundaries are worth stating plainly. First, the 0.601 cage binds, by theorem, every predictor whose channel is (multiset, current move) which is the *position-free* causal single layer and every order-blind mechanism; a positionally encoded causal layer formally sees more, so for it the correct statement is empirical (it measures 0.40 and decays). We write "capped by," never "explained by": the ceiling says why standard attention cannot be good here, not why it attains the particular value it does. Second, under the causal mask the ring is open and the closing edge never enters the output, so this experiment validates the *path-ordered composition* half of the mechanism the transports as ordered memory not loop closure; the load-bearing evidence for holonomy proper remains the ablation of section 6.1. Third, the position-averaged accuracy flatters long sequences (early positions are easy); per-position accuracy decays with depth into the sequence, so the extrapolation is graceful compounding drift, not lossless transfer. Together with section 7, the picture is now two-sided: the loop composition is not discovered from a thin binary label, and the same transport machinery is learned from task labels alone when supervision is dense the regime that sequential prediction, and agentic state tracking, naturally provide.

9 Contextual Inconsistency and Future Directions

The validations in section 6 establish that holonomy is a learnable, recoverable quantity under the proposed parameterization, including in the observable-content regime. The load-bearing ablation of section 6.1 further shows that the loop composition is used by the attention mechanism itself. Neither establishes that any naturally occurring dataset carries above-baseline holonomy. The settings below are therefore presented as *directions for future work* domains in which contextual inconsistency is conceptually natural and where the question of whether the relational structure

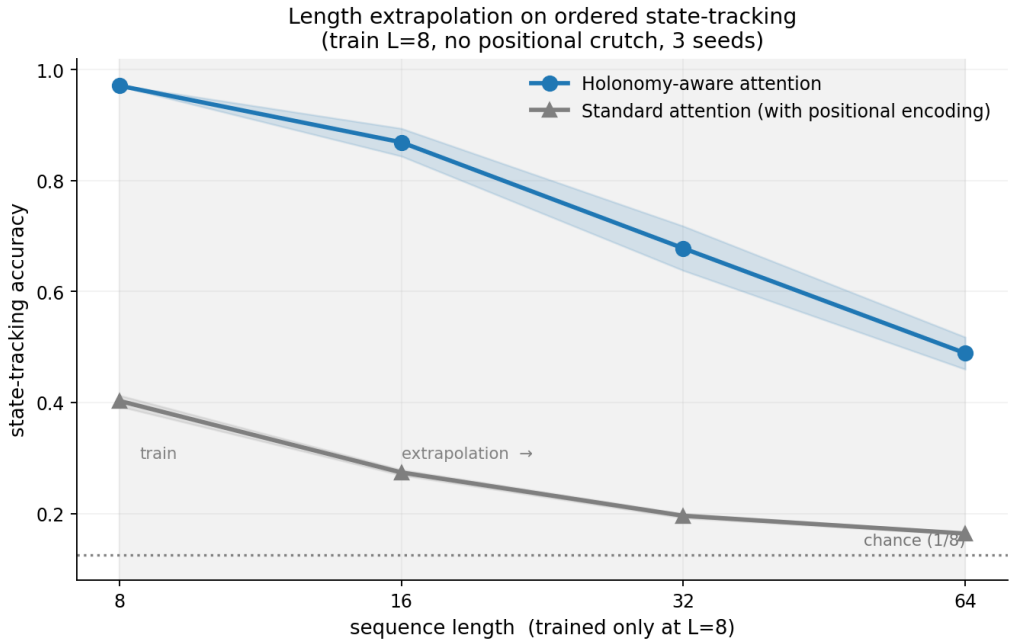


Figure 2: Length extrapolation on ordered state tracking (train $L = 8$, dense per-step labels, no auxiliary loss, 3 seeds). The holonomy-aware layer learns the update rule and transfers far beyond its training length; the positionally encoded standard baseline collapses toward chance. Combined with the exact ceiling of theorem 4 (0.601 for any order-blind predictor granted the current move), the 0.971 at $L = 8$ is an enumerated certificate of order use.

exhibits genuine, above-floor holonomy can be posed rather than as demonstrated applications. In each case the empirical question is open, and our controlled setting suggests the appropriate test: whether a learned connection recovers holonomy exceeding an empirically calibrated noise floor *and* exceeding symmetric baselines.

The proposed framework is motivated by systems in which locally consistent observations cannot be extended to a globally consistent representation. Such situations arise in contextual systems, where different contexts admit internally coherent descriptions but their composition leads to an inconsistency. In the geometric formulation developed in this work, this phenomenon appears as nontrivial holonomy in the transport structure over the token graph.

9.1 Multi-Perspective Systems

Contextual inconsistency naturally arises in systems involving multiple perspectives or agents. Each perspective may provide a locally coherent description of a situation, yet translations between perspectives may introduce transformations that accumulate along chains of interactions. When information propagates through several perspectives and returns to the original one, the resulting representation may differ from the starting representation. Such behavior corresponds to nontrivial holonomy in the underlying transport structure. The load-bearing experiment of section 6.1 is a concrete, controlled instance of this path-dependence.

9.2 Quantum Contextuality

Contextuality also plays a central role in quantum theory. In quantum mechanics, measurement outcomes cannot generally be explained by assigning pre-existing values to observables independently of the measurement context. This phenomenon was formalized by the Bell–Kochen–Specker theorem, which demonstrates the impossibility of non-contextual hidden-variable models for quantum systems of dimension three or greater [12].

Recent work has further explored the relationship between contextuality and other nonclassical phenomena such as quantum nonlocality. For example, Cabello showed that forms of quantum contextuality can be systematically converted into violations of Bell inequalities, highlighting the deep structural connection between these phenomena [6].

The geometric interpretation developed in this paper suggests a possible computational perspective on such systems. In this view, contexts correspond to local representation spaces and compatibility relations correspond to transport operators. Nontrivial holonomy then represents the obstruction to constructing a globally consistent assignment of measurement outcomes.

9.3 Diagnosing Path Dependence in Transformer Systems

Recent work has identified path-dependent behavior in transformer architectures, where model outputs may vary under seemingly innocuous changes such as reordering of context or evaluation order [7]. The WILSON diagnostic framework introduces curvature-inspired probes, based on Wilson-loop–like measurements over internal activations, to detect such order sensitivity and invariance violations in large language models. These diagnostics reveal that transformer computations can exhibit non-commutative effects across positions and layers, leading to practical reliability issues in applications such as retrieval-augmented generation and multi-step reasoning. The discrete bundle perspective proposed in this work offers a complementary viewpoint: path-dependent behavior arises naturally from nontrivial holonomy in the relational transport structure. This suggests that geometric diagnostics of transformer behavior could be interpreted through explicit transport operators and loop compositions within the bundle framework.

9.4 Multi-Hop Relational Reasoning in Structured Data

Another important application domain concerns multi-hop relational reasoning over structured data such as knowledge graphs. Recent work has shown that standard transformer architectures face theoretical limitations when performing multi-hop reasoning, requiring depth proportional to the number of reasoning steps [14]. Approaches such as Relation-Aware Sparse Attention introduce structural inductive biases by restricting attention to graph edges and incorporating relation-specific attention preferences. From the perspective of the present framework, relational reasoning can be interpreted geometrically through transport along paths in the token interaction graph. Compositions of transport operators along multi-step relational chains naturally encode path structure, suggesting that discrete bundle connections may provide an alternative mechanism for modeling multi-hop relational dependencies within attention-based architectures.

9.5 Probabilistic and Financial Contextuality, and the Question of Observables

A further direction concerns probabilistic and financial systems, where global consistency defects admit a transport interpretation. In the categorical framework of Adachi and collaborators, conditional-expectation operators over a filtration play the role of transport between probabilistic contexts, and an Aharonov–Bohm–type arbitrage appears as a holonomy-like obstruction to globally

trivializing these transports [4]. The learned-connection viewpoint developed here is complementary: rather than specifying the transport categorically, one would attempt to *infer* an effective connection from observed dynamics and test whether its holonomy exceeds the noise-induced floor.

We emphasize that this is an open empirical question, and a subtle one. Whether financially or probabilistically meaningful non-trivial holonomy can be robustly detected is, at present, unresolved. Two cautions are worth stating explicitly. First, for holonomy to be a property of the data rather than of an imposed construction, the underlying states must carry a genuine internal degree of freedom a fiber relative to which transport can act; where no such intrinsic degree of freedom exists, any computed holonomy reflects the modeling choice rather than the system. Second, the relevant structure, if present, may not reside in the most readily available observables: pairwise correlations of coarse time series may average away the very inconsistency one seeks, and higher-order relations, microstructure, or multi-scale information flows may be more natural carriers. Distinguishing genuine, above-floor holonomy from imposed or noise-induced structure in such settings is the central difficulty, and we regard it as the key open problem for any real-world instantiation of this framework. The trainability limitation of section 7 is a concrete reminder that representability and recoverability under supervision do not by themselves imply that the structure can be induced from task labels alone.

Contextuality originates in the foundations of quantum mechanics, where it captures the incompatibility between local measurement assignments and globally consistent hidden-variable models. In the sheaf-theoretic formulation introduced by Abramsky and Brandenburger [2], contextuality arises when locally consistent assignments of measurement outcomes cannot be extended to a single globally consistent assignment across all contexts. Contexts correspond to sets of jointly measurable observables, and contextuality appears as an obstruction to constructing a global section of the associated presheaf.

This formulation provides a structural perspective on contextuality that is independent of the specific physical realization of the measurements. Subsequent work has further developed this viewpoint using tools from algebraic topology, such as cohomological invariants that detect contextuality as a global obstruction [3]. In this sense, contextuality can be interpreted broadly as the failure of locally defined relations to assemble into a globally consistent structure. In the abelian case, this obstruction is classified by H^1 of the associated sheaf of local sections [3]; our holonomy defect $\mathrm{inc}(\gamma)$ provides a directly computable realization of the same obstruction in the non-abelian $\mathrm{SO}(d)$ setting, where the matrix norm replaces the abstract cohomological invariant.

10 Related Work

10.1 Holonomic and gauge-constrained sequence models

Closest in mathematical machinery to the present work is the recent Holonomic Network of Sung [18], which represents a recurrent hidden state's evolution as a path-ordered product of input-conditioned $\mathrm{SO}(N)$ operators and argues that the resulting non-abelian holonomy yields a "symmetry-protected" memory robust to noise, with strong length generalization on symbolic reasoning tasks. We share the core mechanism skew-generated $\mathrm{SO}(N)$ transports composed multiplicatively but differ in object and purpose. Sung's holonomy is computed *along the time axis of a single model*, where it serves as an *architectural* mechanism that protects sequential memory; our holonomy is computed *across a relational graph connecting distinct representations*, where it serves as a *diagnostic* of contextual inconsistency between them. Consequently the two are complementary rather than competing: one asks how to make a single computation robust, the other asks how to measure whether a collection of local viewpoints can be globally reconciled. The telescoping obstruction of section 4.1 that

inconsistency must reside in relations rather than node content has no analogue in the single-stream setting and is specific to the relational reading pursued here.

10.2 Geometric and Gauge-Equivariant Neural Networks

A closely related line of research in geometric deep learning studies neural architectures that are equivariant with respect to local frame transformations, often referred to as gauge transformations. Gauge equivariant convolutional networks construct feature representations that transform consistently under changes of local coordinate frames on manifolds or grids [8]. These models ensure that convolutional operations respect underlying symmetry structures induced by the geometry of the domain.

While these approaches also introduce the notion of local frames and gauge transformations, their goal is typically to enforce equivariance of neural computations under prescribed symmetry groups. In contrast, the present work focuses on learning a discrete connection over a token graph and explicitly performing parallel transport between local representation spaces. Rather than imposing symmetry constraints, the learned transport operators may induce nontrivial holonomy, which we interpret as a signal of contextual inconsistency in relational representation learning.

10.3 Geometric and Manifold-Based Attention Mechanisms

Recent work has also explored geometric interpretations of attention mechanisms. For example, RiemannFormer models attention interactions on a Riemannian manifold, treating queries and keys as tangent vectors and defining similarity through the manifold metric with parallel transport between tangent spaces [11]. In contrast, our framework operates on discrete bundle structures defined over token interaction graphs and studies learned transport operators whose composition can produce nontrivial holonomy.

10.4 Discrete transport and parallel transport on graphs

Closely related geometric ideas appear in the study of parallel transport on graphs and manifolds. Vector diffusion maps [17] introduce a framework in which edges of a graph carry linear operators representing parallel transport between local vector spaces attached to each node. In this setting, compositions of transport operators along cycles naturally define holonomy, providing a measure of the curvature of the underlying connection.

This approach has been used to analyze geometric structure in data by constructing connection Laplacians that capture both diffusion and transport effects across a graph. The resulting framework provides a discrete analogue of geometric parallel transport and has been applied to problems involving alignment, synchronization, and manifold learning [17, 10]. These constructions demonstrate how nontrivial holonomy can arise in discrete systems when locally defined transports fail to compose consistently around cycles.

10.5 Vector Bundle Attention

Recent work has explored incorporating vector bundle structures into transformer architectures by aligning feature vectors through learned parallel transport before similarity computation. For example, the Vector Bundle Attention framework introduces a geometry-aware attention mechanism in which fiber features are aligned using endpoint-conditioned transport operators prior to computing attention scores [9]. In contrast, the present formulation operates on discrete bundle structures

defined over token interaction graphs and explicitly models the effects of transport composition, allowing nontrivial holonomy to emerge from relational cycles.

This formulation allows the relational geometry between tokens to influence attention computation. As we show in the preceding sections, nontrivial holonomy in the connection introduces path-dependent effects, enabling context-sensitive relational comparisons.

10.6 Sheaf-Based Neural Networks

Another closely related line of work models graph-structured data using sheaf-theoretic constructions. Sheaf neural networks associate vector spaces with graph nodes and linear maps with edges, enabling the representation of locally varying feature spaces.

While these models employ similar mathematical structures, they are typically used to define diffusion processes or generalized Laplacian operators on graphs. In contrast, the present work integrates transport operators directly into the attention mechanism, allowing relational comparisons to depend on the geometry induced by the learned connection.

10.7 Holonomy and contextuality in discrete bundle models

Recent work has explored the relationship between contextuality and geometric structures on graphs. In particular, Magnot [13] introduced a framework in which relational weights take values in a group G and define a discrete bundle over a graph. In this formulation, edge weights act as transition functions, and transport along a path is given by the ordered product of group-valued weights. Holonomy computed around cycles provides a measure of global inconsistency in the relational structure, and contextuality corresponds to the failure of the system to admit a globally consistent trivialization.

While this framework studies the geometry of relational transformations themselves, the present work focuses on relational comparison between vector representations. Instead of associating group elements directly with nodes or relations, we attach vector spaces to tokens and introduce transport operators that align representations between local fibers. Holonomy in this setting measures contextual inconsistency in relational comparisons between token representations rather than inconsistency in the transformation network itself.

11 Discussion

The present work focuses on orthogonal transport operators $U_{ij} \in SO(d)$ in order to preserve inner products and stabilize relational comparisons. However, the geometric formulation introduced here naturally extends to other Lie groups.

In particular, different applications may benefit from transport operators belonging to alternative structure groups, such as $U(d)$, $GL(d)$, or more specialized groups reflecting symmetries present in the underlying data. Exploring such generalizations may allow the model to capture richer relational transformations, including non-orthogonal or complex-valued representations.

This perspective follows earlier work connecting contextual systems and learning dynamics with group-valued transport structures, including the framework proposed by Magnot.

12 Conclusion

We introduced a geometric formulation of attention in which token representations are modeled as elements of local vector spaces connected by learned transport operators. By replacing global

alignment with parallel transport, relational comparisons become dependent on the geometry of the underlying connection.

This perspective naturally leads to a discrete fiber bundle structure over the token graph, where holonomy provides a quantitative measure of contextual inconsistency and path dependence. We established the basic theoretical properties of the resulting holonomy-aware attention mechanism exact reduction to standard attention under flat connections, gauge invariance, and path dependence under nontrivial holonomy and we proved a representational lower bound showing that a single standard attention layer with a symmetric read-out cannot represent an ordered-product (holonomy) label. On a shortcut-free instance of such a label, a matched-parameter ablation shows that loop closure is load-bearing within the attention layer itself, with its contribution growing as the learned transports become accurate.

We further showed, via a parameter-matched ablation within the attention layer, that the loop composition is load-bearing a causal arm consistently outperforms an otherwise identical inert arm, with the gap growing as the learned transports become accurate while being explicit that this holds under transport-level supervision and that the same structure is not discovered from task labels alone.

These results establish holonomy as a learnable and recoverable quantity under the proposed parameterization, and frame attention as a geometric object in which consistency and context sensitivity arise from properties of the connection. They also delimit what has *not* been shown: we make no claim that naturally occurring data carries non-trivial, above-baseline holonomy. The central open problem for any real-world instantiation is to identify domains and observables in which genuine, above-floor holonomy is present, and to distinguish it from imposed, symmetric, or noise-induced structure. Further directions include learning richer structure groups, incorporating curvature regularization, end-to-end trainability of the attention layer without transport supervision, and connecting the framework to related group-valued and sheaf-theoretic models of contextuality.

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A Proofs for the Representational Results (section 6.1)

Setup. Fiber dimension $d \geq 3$; a ring of L tokens. A sample plants edge rotations $G_0, \ldots, G_{L-1} \in \mathrm{SO}(d)$: the first $L-1$ (the *free* edges) are drawn i.i.d. from a fixed distribution independent of the label, and the closing edge is set to $G_{L-1} = \text{target} \cdot (G_{L-2} \cdots G_0)^{-1}$ with target $= I$ (label 0) or a fixed π-rotation R_π (label 1), so the ordered product is $H = G_{L-1} \cdots G_0 = \text{target}$. Token t is a fixed injective linear encoding of $\mathrm{vec}(G_t)$; the formal results below concern the position-free read-out (residual add, mean-pool over positions, linear classifier), and the positionally encoded baseline is assessed empirically in the main text.

Proof of lemma 1. Write $X' = \sigma \cdot X$ for the permuted input, so $x'_t = x_{\sigma^{-1}(t)}$. Queries, keys and values relabel: $q'_t = q_{\sigma^{-1}(t)}$, and likewise k', v'. Scores relabel in both indices: $s'_{ij} = s_{\sigma^{-1}(i)\,\sigma^{-1}(j)}$. The softmax at position i normalizes over a full row, and a sum over all entries of a row is unchanged by reordering its entries, so $\alpha'_{ij} = \alpha_{\sigma^{-1}(i)\,\sigma^{-1}(j)}$. Substituting $j'' = \sigma^{-1}(j)$ in $o'_i = \sum_j \alpha'_{ij} v'_j$ gives $o'_i = o_{\sigma^{-1}(i)}$, hence $h'_i = h_{\sigma^{-1}(i)}$ after the residual add. Mean-pooling sums over all positions and is therefore unchanged: $\frac{1}{L}\sum_i h'_i = \frac{1}{L}\sum_i h_i$. A linear classifier of an unchanged vector is unchanged. Multiple heads satisfy the argument headwise. Hence the decision function is invariant under every permutation, at every parameter setting. □

Proof of proposition 2. Let $A, B \in \mathrm{SO}(d)$ be non-commuting (for $d \geq 3$ take $90°$ rotations in the $(1,2)$ and $(2,3)$ planes; $AB \neq BA$ by direct computation). Set $G_0 = A$, $G_1 = B$, $G_2 = \cdots = G_{L-2} = I$, and $G_{L-1} = R_\pi (BA)^{-1}$. Then $H = R_\pi (BA)^{-1} BA = R_\pi$, so $\hat{y} = 1$. Swap tokens 0 and 1: the product becomes $H' = R_\pi (BA)^{-1} AB$, and $H' = R_\pi$ would force $AB = BA$, a contradiction; so $\hat{y} = 0$ on the swapped input. By lemma 1 any function in the class takes equal values on the two inputs, so none equals $\hat{y}$. □

Remark 7 (What invariance does not give: the on-support ceiling)**.** Proposition 2 rules out equality with $\hat{y}$ on all of $\mathrm{SO}(d)^L$; it does *not* bound accuracy on the task distribution. On the support, the unordered multiset of edges almost surely determines the label: test all $L!$ orderings and check whether some ordered product has the eigenvalue fingerprint of a π-rotation (two eigenvalues -1). On class 1 the planted ordering yields R_π exactly (and cyclic shifts yield conjugates $QR_\pi Q^{-1}$, with the same fingerprint); on class 0 a generic re-ordering of generic rotations hits a π-rotation only on a measure-zero set. This ordering search is permutation-invariant and classifies perfectly almost surely, so the invariant-class ceiling on this task is 1, not $1/2$. Accuracy control must therefore come from elsewhere: theorem 5 below for every partial-view strategy, and measurement for the full-view baseline.

Subset blindness under Haar edges. Assume now
(H) the free edges $G_0, \ldots, G_{L-2}$ are i.i.d. Haar on $\mathrm{SO}(d)$,
and the classes are equiprobable. Recall the defining invariances of the (unique) Haar probability measure on the compact group $\mathrm{SO}(d)$: if $G \sim$ Haar and R is fixed, then $RG \sim$ Haar and $GR \sim$ Haar, and $G^{-1} \sim$ Haar.

Lemma 2 (Haar randomization)**.** *Let $G \sim$ Haar be independent of a random $X \in \mathrm{SO}(d)$ (arbitrary law). Then $XG \sim$ Haar and XG is independent of X; likewise for GX.*

Proof. Conditionally on $X = x$, independence gives $G \sim$ Haar, so $xG \sim$ Haar by left invariance. The conditional law of XG given X is thus Haar for every value of X: this is independence, and averaging over X gives the unconditional law Haar. □

Theorem 5 (Subset blindness)**.** *Under* (H)*, for every strict subset $S \subsetneq \{0, \ldots, L-1\}$ of the edge positions, the joint law of $(G_t)_{t \in S}$ is identical under the two classes.*

Proof. If $L - 1 \notin S$, every edge in S is free and the free edges are generated without reference to the target; their joint law is $\mathrm{Haar}^{|S|}$ under both classes. Otherwise S omits some free edge G_j $(0 \leq j \leq L-2)$. Split the free product around j: $G_{L-2} \cdots G_0 = A\, G_j\, B$ with $A = G_{L-2} \cdots G_{j+1}$ and $B = G_{j-1} \cdots G_0$ built from the other free edges (empty products are I). Then

$$G_{L-1} = \mathrm{target} \cdot (A\, G_j\, B)^{-1} = \mathrm{target} \cdot B^{-1} G_j^{-1} A^{-1}.$$

Condition on all free edges except G_j; this fixes $A = a$, $B = b$, and every free edge in S. Conditionally, $G_j \sim$ Haar, so $G_j^{-1} \sim$ Haar, so $b^{-1} G_j^{-1} \sim$ Haar (left invariance), so $b^{-1} G_j^{-1} a^{-1} \sim$ Haar (right invariance), and finally $\mathrm{target} \cdot b^{-1} G_j^{-1} a^{-1} \sim$ Haar (left invariance) for *either* value of the target. Hence the conditional law of G_{L-1} given the observed free edges is Haar under both classes, and the marginal law of the observed free edges is class-independent; the joint law of $(G_t)_{t \in S}$, being determined by these two ingredients, is identical under the two classes. □

Corollary 1 (Every subset read-out is exactly at chance)**.** *Under* (H) *with equiprobable classes, any classifier, deterministic or randomized, whose output depends on the input only through $(G_t)_{t \in S}$ for some strict subset S, has accuracy exactly $1/2$.*

Proof. Let $p = \Pr(\phi = 0 \mid y = 0)$. By theorem 5 the classifier's input has the same law under both classes, so its (possibly randomized) output does too, and $\Pr(\phi = 0 \mid y = 1) = p$. Then $\mathrm{Acc} = \frac{1}{2}p + \frac{1}{2}(1 - p) = \frac{1}{2}$. □

Remark 8 (Strictness and the edge distribution)**.** Strictness is necessary: the full set of edges determines the label exactly (the ordered product is I or R_π). The Haar assumption is likewise doing real work: for a rigid edge distribution (e.g., supported on two rotations swapped by the target), the closing edge alone leaks the label, even though every edge is "random." Uniformity in the Haar sense is exactly the shift-invariance that erases the target's fingerprint from every partial view. The experiments in the main text draw free edges as exp of a scaled random skew matrix, which is close to but not exactly Haar; the theorem is stated for the Haar variant, a one-line change to the generator.

Multi-layer case (conditional). A deep transformer with MLPs can compose pairwise interactions across depth, so proposition 2 does not assert "no transformer can." Representing $\hat{y}$ requires either depth growing with L (iteratively composing the order-L product) or a latent state large enough to carry the ordered partial products. Encoder–decoder architectures, including Transformers, fall within the locally-Lipschitz class of Anschuetz et al. [5] (their Theorem 3), with effective memory counted as model dimension times sequence length; the latent-memory route therefore incurs their memory lower bound. Their separation is worst-case (existence of inputs forcing an error); theorems 4 and 5 are distributional; neither implies the other. We state this as a conditional and do not rely on it for the main claim.

B Proof of the State-Tracking Ceiling (theorem 4)

Proof of theorem 4. Condition on the summary value $T = \tau$. The predictor's output on this event is a (possibly random) label depending only on τ; write $q_\tau(y)$ for the probability it answers y, with $\sum_y q_\tau(y) = 1$. Since the answer depends only on τ, it is conditionally independent of Y_t given $T = \tau$, so

$$\Pr(\text{correct} \mid T = \tau) = \sum_y q_\tau(y) \Pr(Y_t = y \mid T = \tau) \le \max_y \Pr(Y_t = y \mid T = \tau),$$

a weighted average with nonnegative weights summing to one cannot exceed its largest term, with equality when all mass sits on an argmax label (the majority vote). Averaging over the law of T (total probability) gives $\mathrm{Acc} \le \mathbb{E}_T[\max_y \Pr(Y_t = y \mid T)]$. With a finite alphabet the right side is computed exactly: enumerate all 4^t prefixes, group by T, take each group's majority fraction, and average. □

Enumerated values. For the task of section 8 (dense metric, mean over positions 1..8, all 4^8 sequences): T = multiset gives 0.488; T = (multiset, current move) gives 0.601; the trivial summary (T constant) gives the majority-class floor 0.214. The per-position ceilings decay from 1.0 at $t \le 2$ (where the summary still determines the prefix) to 0.26–0.29 at $t = 8$. The buckets are not uniform at $t = 1$ only three buckets are reachable and the largest carries half the mass which is why the majority floor exceeds the naïve 1/8.